GHAZI UNIVERSITY

LAB REPORT

Programming Fundamentals Lab 03-04

Topic: Variables, Expressions, Type Casting & Math Tutor Program

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Date: 15/10/26

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Task 1

Assume that the following variable are defined:

```
int age;
double Pay;
char Section;
```

write a Single cin Statement that will read input in to each of these variables.

[input]

```
#include <iostream>
using namespace std;
int main()
{
 int age;
 double Pay;
 char Section;
 cout << "Enter age, Pay and section: ";
 cin >> age >> Pay >> Section;
 cout << "Age:" << age << endl;
 cout << "Pay:" << Pay << endl;
 cout << "Section:" << Section << endl;
 return 0;
}
```

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[output]

Enter age, Pay and Section: 18 18000 C

Age = 18

Pay = 18000

Section = C

new*

```cpp
#include <iostream>
using namespace std;
int main()
{
    int age;
    double Pay;
    char Section;
    cout << "Enter age, Pay and section: ";
    cin >> age >> Pay >> Section;
    cout << "Age = " << age << endl;
    cout << "Pay = " << Pay << endl;
    cout << "Section = " << Section << endl;
    return 0;
}
```


TAB
Enter age, Pay and section: 18
18000
c
Age = 18
Pay = 18000
Section = c
[Program finished]

Task #2

complete the table by writing the value of each expression in the value column according to c++ language rule.

```
##include <iostream>
using namespace std;
int main () {
cout << 28/4-2 << endl;
cout << 6+12*2-8 << endl;
cout << 6+17 %3-2 << endl;
cout << 2+22*(9-7) << endl;
cout << (8+7)*2 << endl;
cout << (16+7) %2-1 << endl;
cout << 12/(10-6) << endl;
cout << (19-3)*(2+2)/4 << endl;
cout << 4+8*2 << endl;
} return 0;
```

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[output]

5

22

6

46

30

0

3

16

20

new*

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << 28/4-2 << endl;
    cout << 6+12*2-8 << endl;
    cout << 6+17%3-2 << endl;
    cout << 2+22*(9-7) << endl;
    cout << (8+7)*2 << endl;
    cout << (16+7)%2-1 << endl;
    cout << 12/(10-6) << endl;
    cout << (19-3)*(2+2)/4 << endl;
    cout << 4+8*2 << endl;
    return 0;
}
```


TAB
5
22
6
46
30
0
3
16
20
[Program finished]

Task #3

Assume int units, float mass, double weight = mass * units. which automatic data type conversion will take place?

```
#include <iostream>
using namespace std;
int main ()
{
    int units = 5;
    float mass = 2.5;
    double weight;
    weight = mass x units;
    cout << "units = " << units << endl;
    cout << "Mass = " << mass << endl;
```

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```
cout << weight = " << weight << endl;
return 0;
}
```

~~input~~]

[output]

units = s

Mass = 2.5

weight = 12.5

new*

```cpp
#include <iostream>
using namespace std;
int main()
{
    int units = 5;
    float mass = 2.5;
    double weight;
    weight = mass * units;
    cout << "units = " << units << endl;
    cout << "Mass = " << mass << endl;
    cout << "weight = " << weight << endl;
    return 0;
}
```


TAB
units = 5
Mass = 2.5
weight = 12.5
[Program finished]

LAB #4

Assume a program has the following variable definitions

int a, b = 2; Float c = 4.2;

and the following statement a = b x c;

what value will be Stored in a?

A. 8.4 B 8 C 0 D none of above

```
#include <iostream>
using namespace std;
int main ()
{
  in a, b = 2;
  float c = 4.2;
  a = b x c; // 2x 4.8 = 8.4 => truncated to 8
  cout << "a = " << a; // output: 8
```

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```
    return 0;
}
```

output, a = 8

new*

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b = 2;
    float c = 4.2;

    a = b * c; // 2 x 4.2 = 8.4 => truncated to 8

    cout << "a = " << a; // output: 8

    return 0;
}
```


TAB
a = 8
[Program finished]

Task. 5

Assume that qty and SalesReps are both integers. use a type cast expression to rewrite the following statement so it will no longer perform integer divison

```
unitsEach = qty/SalesReps;
```

```
#include <iostream>
using namespace std;
int main()
{
    int qty = 10;
    int SalesReps = 3;
    double unitsEach;
unitsEach = qty/salesReps;
cout << "without casting(integer Divison): "
        << unitsEach << endL;
unitsEach = Static_cast<double>(qty)
        SalesReps;
cout << "with casting (correct Divison):"
```

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```
    << unitsEach << endl;
    return 0;
}
```

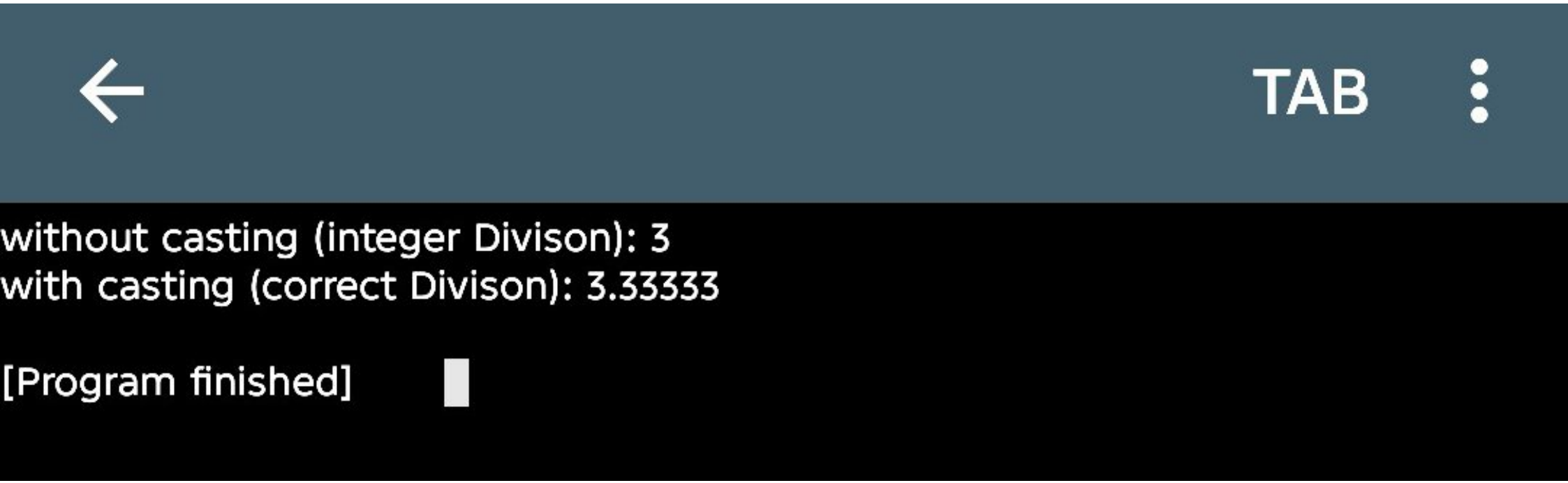
[output]

without casting (integer Divison): 3

with casting (correct Divison): 3.33333

new*

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      int qty = 10;
6      int SalesReps = 3;
7      double unitsEach;
8
9      unitsEach = qty / SalesReps;
10     cout << "without casting (integer
   Divison): "
11         << unitsEach << endl;
12
13     unitsEach = static_cast<double>(qty) /
   SalesReps;
14     cout << "with casting (correct Divison):
   "
15         << unitsEach << endl;
16
17     return 0;
18 }
```


TAB
without casting (integer Divison): 3
with casting (correct Divison): 3.33333
[Program finished]

Task - 6

write a program that can be used as a math tutor such as 247 + 129

```
#include <iostream>
#include <cstdlib>
//int main ()
{
  srand (time(0));
 int num1 = rand() % 900 + 100;
 int num2 = rand() % 900 + 100;
 cout << " " << num1 << endl;
 cout << "+" << num2 << endl;
 cout << "----" << endl;
 cout << "\npress Enter to see answer";
 cin.get();
```

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```
    cout << "\n" << num1 << endl;
    cout << "+" << num2 << endl;
    cout << "----" << endl;
    cout << " " << num1+num2 <<<< endl;
    return 0;
}
```

[output]

```
  515
+ 532
-----
 515
+532
-----
1047
```

new*

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

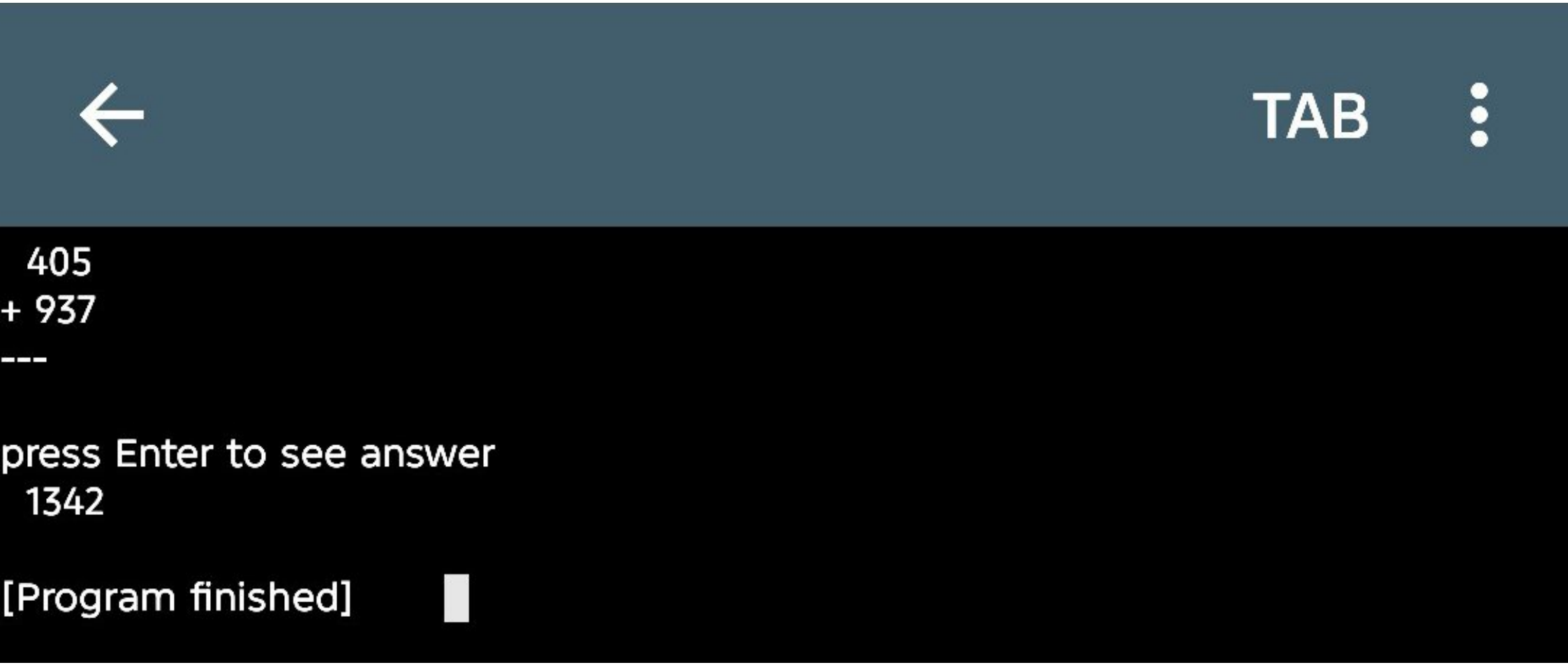
int main()
{
    srand(time(0));
    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "---" << endl;

    cout << "\npress Enter to see answer";
    cin.get();

    cout << "  " << num1 + num2 << endl;

    return 0;
}
```


TAB
405
+ 937
---
press Enter to see answer
1342
[Program finished]

[Home task]

Task 1

Locate error

program-1

```
using namespace std;
void main()
{
double number1, number2, Sum;
cout << "Enter a number: ";
cin >> number1;
cout << "Enter another number: ";
```

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```
cin << number 2;
number 1 + number 2 = Sum;
cout "The Sum of the two numbers is"
                <<Sum
}
```

[errors]

1 [missing header]: # include <iostream> is missing at the top.

2 [Main function:] Void main() is wrong it should be int main().

3 [capital C:] cin << number; is wrong Cin Should be Small cin and <<Should be >> for input. correct: cin >> number1;

4 [wrong operator:] cin << number2; is wrong.

5 [wrong Assignment:] number1 + number2 = Sum is wrong.

6 [Missing operator:] cout " The Sum..." is missing << after cout. correct: cout << " The Sum..."

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Program 2 (error)

```
#include <iostream>
using namespace std;
void main ()
{
 int number1, number2;
 float quotient;
 cout << "Enter two number and I will
          divide \n";
 cout << "the first by the second for you.\n";
 cin >> number1, number2;
 quotient : float <Static_cast>(number1)/
     number2;
 cout << quotient.
```

[Error]

- void main() Should be int main()
- cin>>number1, number2; Should be
  cin>> number1 >> number2;
- float <Static_cast>(number1) → wrong Syntax
  Should be Static_cast<float>(number1)
- cout << quotient → missing Semicolon;

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Task-2

Average of values to get the average of a series of values, add them and divide the sum by number of values. write a program that stores the following values in five different variable 26, 32, 37, 24, and 33.

Program

```
#include <iostream>
using namespace std;
int main()
{
 int a=26, b=32, c=37, d=24, e=33;
 int Sum;
 float average;
 Sum = a+b+c+d+e;
 average = Sum/5.0;
 cout << "Sum = " << Sum << endl;
 cout << "Average = " << average;
 return 0;
}
```

output

Sum = 157

Average = 30.8

new*

```cpp
#include <iostream>
using namespace std;
int main()
{
    int a = 28, b = 32, c = 37, d = 24, e = 33;
    int Sum;
    float average;
    Sum = a + b + c + d + e;
    average = Sum / 5.0;
    cout << "Sum = " << Sum << endl;
    cout << "Average = " << average;
    return 0;
}
```


TAB
Sum = 154
Average = 30.8
[Program finished]

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LAB - 4

Task-1

write a program to print positive number entered by the user. if the user enters negative number print number entered is positive otherwise print number is negative.

```
#include <iostream>
using namespace std;
int main()
{
  int n;
  cout << "Enter a number:";
  cin >> n;
 if (n>=0)
 cout << "Positive";
 else
 cout << "Negative";
  return 0;
 }
```

Enter a number : 8
Positive

new*

```cpp
#include <iostream>
using namespace std;
int main()
{
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n >= 0)
    cout << "Positive";
    else
    cout << "Negative";
    return 0;
}
```


TAB
Enter a number: 8
Positive
[Program finished]

Day: Date:

Task-2

```
#include <iostream>
using namespace std;
int main()
{
 int num;
 cout << "Enter an integer:";
 cin >> num;
 if (num > 0)
 cout << "Number is positive";
 else if (num < 0)
 cout << "Number is negative";
 else
 cout << "Number is zero";
 return 0;
}
```

out put

```
Enter an integer: 7
Number is positive
```




```cpp
#include <iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter an integer: ";
    cin >> num;
    if (num > 0)
    cout << "Number is positive!";
    else if (num < 0)
    cout << "Number is negative!";
    else
    cout << "Number is zero!";
    return 0;
}
```


TAB
Enter an integer: 7
Number is positive!
[Program finished]

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Task - 3

```
#include <iostream>
using namespace std;
int main()
{
 int marks;
 cout << "Enter Marks: ";
 cin >> mark;
 if (mark > 100 || mark < 0)
 {
  cout << "invalid Marks!";
 }
 else if (mark >= 75)
 {
  cout << "Grade A";
 }
 else if (mark >= 60)
 {
  cout << "Grade B";
 }
  else if (mark >= 45)
 {
```

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```
cout << "Grade c";
}
    return 0;
}
```

Enter mark: 1200

invalid marks

new*

```cpp
#include <iostream>
using namespace std;
int main()
{
    int marks;
    cout << "Enter Marks: ";
    cin >> marks;

    if (marks > 100 || marks < 0)
    {
        cout << "invalid Marks!";
    }
    else if (marks >= 75)
    {
        cout << "Grade A";
    }
    else if (marks >= 60)
    {
        cout << "Grade B";
    }
    else if (marks >= 45)
    {
        cout << "Grade C";
    }
    else if (marks >= 33)
    {
        cout << "Grade D";
    }
    else
    {
        cout << "Fail";
    }
    return 0;
}
```


TAB
Enter Marks: 1200
invalid Marks!
[Program finished]

Shahmeer

Task # 4

```
#include <iostream>
using namespace std;
int main ()
{
  int n, c = 0;
  cout << "Enter Number: ";
  cin >> n;
 if (n>0)
{
  while (n>0) {c ++; n/=10;}
  cout << "Digits: " << c;
 } else {
        cout << "Not Positive";
        }
 }
```

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output Enter a number : 5

Digit : 1

new*

```cpp
#include <iostream>
using namespace std;
int main()
{
    int n, c = 0;
    cout << "Enter Number: ";
    cin >> n;

    if (n > 0)
    {
        while (n > 0)
        {
            c++;
            n /= 10;
        }
        cout << "Digits: " << c;
    }
    else
    {
        cout << "Not Positive";
    }

    return 0;
}
```


TAB
Enter Number: 5
Digits: 1
[Program finished]